Elvis A. F. Martis

A Practical Guide to Molecular Docking

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Introduction

Drug discovery[**?**] is an arduous journey with many unknowns throughout the path, with several traps that lead to failure. To perform a large number of trials in an iterative manner, computer simulations[**?**, **?**] are convenient to perform tens of thousands of calculations to provide crucial insights, and possibly, rule out dead ends. It is always said that drug discovery is a multi-faceted science that also requires a great deal of art and innovative thinking. So how do we train computers to think like humans? This is never a worry because we do not need to train computers to think like humans. As a matter of fact, computer are extremely fast but dumb machines. Whereas humans are extremely slow, lazy but smart thinkers. Now, the division of labour between humans and computers should be straightforward; let the computers do the repetitive tasks such as calculating several mathematical equations that can be programmed to describe chemistry and biology. This is done employing the laws of physics that govern the movement of the atoms, and therefore molecules. Humans can engross themselves in decision-making that is beyond any mathematical equations. In the recent times with generative artificial intelligence (**genAI**), computers can perform some level of innovative thinking, however, their chemical and biological validity, in a general sense, remains to be tested. GenAI, therefore, deserves an appreciation to at least recognise the idea that computers could be made to think, not in biological sense, rather in mathematical sense that is converted to chemical or biological sense. However, the later part has been extremely challenging and yet to be conquered.

Structure-based drug discovery (SBDD) is one such approach that aids in designing and optimizing chemical entities by leveraging the tertiary structural information of biological targets. SBDD encompasses various strategies, including virtual screening, where large libraries of compounds are screened for its binding affinity with the target site. This process is also called as “hit identification”, where hits are referred to as molecules that show weak yet measurable binding affinity. Hits generally contain most of the chemical framework capable of showing the desired pharmacological response, however, they need optimisation to improve the binding affinity, and other physico-chemical properties essential for developing a clinical candidate. Most of the docking programs have shown great success at the lead modification stage, which involves evaluating the effect of the new substituents on the same central core (also called a parent nucleus)[**?**].

Another important method, called as *de novo* drug design, involves constructing new molecules from molecular fragments within the binding site of the target. Molecular docking[**?**] is an important technique in structure-based drug discovery (SBDD) that predicts optimal binding conformation of a small molecule (ligand) within the binding site of a biological target. This process helps identify potential drug candidates by estimating their binding affinity and types of interactions to receptor atoms. The latter information is used to suggest modifications in the hit molecules, such that their binding affinity is improved. Iterative cycles of design, synthesis, and experimental validation refine these compounds to optimize their pharmacological properties, this process is called as "lead optimisation".

With a near exponential increase in the tertiary structure information in the last 3 years attributed to several artificial intelligence based methods, particularly Alphafold[**?**] and RoseTTAFold[**?**], many efforts have been made to incorporate machine learning methods in drug discovery to improve the quality of therapeutic agents and reduce the drug discovery timelines significantly[**?**]. Furthermore, developers are also constantly developing methods that employ machine learning methods to improve the prediction accuracy of molecular docking methods. Incorporating the state-of-the-art deep learning neural networks have shown to improve conformational search algorithms and develop better and more generalised scoring functions. These improvements have helped ameliorate challenges posed by limited structural and small molecule data[**?**, **?**].

Molecular docking has been used for more than four decades that serves to attain two primary objectives:

- To predict the binding affinity and conformation of small molecules within a receptor site, and
- To identify hits from a large chemical database to search for diverse chemical scaffolds.[**?**].

Despite classifying these objectives behind molecular docking into separate classes, their boundaries are not clearly demarcated. To attain these objectives, molecular docking programs must comprise components that can search the conformational space within the binding site and a scoring function that scores the conformations such that the biological relevant ones are ranked higher.

Conformational Search Methods

One of the most challenging aspect of pose prediction is the conformational search algorithm. In general, there are two main classes of conformational search methods, systematic and stochastic. The users do not always have a choice to select the search algorithm because the developers usually select based on their expertise and code development. Nevertheless, there are programs, such as AutoDock[**?**], that allow users to select the search algorithm. Here, we succinctly discuss commonly used conformational search methods, however, the list is far from being comprehensive, interested readers are directed to consult the following papers for detailed discussion[**?**, **?**].

Systematic Methods:

These methods thoroughly explore all potential conformations exhaustively by systematically changing one or more torsional degrees of freedom. These degrees of freedom arise from the single bonds that can be rotated to attain different atomic orientation with respect to each other (henceforth called as "rotatable bonds"). We present a brief outline of the methods that fall within the realm of systematic search methods.

- *Systematic Search*: The systematic search algorithm[**?**] rotates all possible rotatable bonds (single bonds in the ring structures cannot be rotated as freely as open-chain structures) systematically by a fixed interval to explore all possible conformations. Such a method can explore all possible conformations, however, the complexity increases exponentially as the number of rotatable bonds increases. There are pruning algorithms that act as a "bump check" that rule out torsion angles that lead to significant atomic overlaps. Furthermore, docking algorithms employ the search algorithm within the binding pocket, and hece, more rotations will be pruned out that may lead to atomic overlaps with receptor atoms. Glide[**?**] and FRED[**?**] are examples of two docking programs that employ this method for conformation search.
- *Incremental Construction*: In this method[**?**], the molecule is broken down into individual fragments. The fragments are then docked in the most suitable sub-pocket in the binding site. The entire molecule is then sequentially built by adding the linkers. This follows a systematic search for the linker conformation that optimally holds the

fragments in their respective sub-pockets. The fragmentation method selects the rigid molecular components, for example ring structures, as fragments, and the flexible components as linkers. In doing so, this method reduces the computational complexity compared to the systematic search by focusing on small components, more particularly the linkers. FlexX[**?**] and DOCK[**?**] are examples of the docking algorithms that employ this method as its conformational search algorithm.

Stochastic Methods

Stochastic techniques utilize random sampling and probabilistic methods to explore the conformational space. They work by randomly making changes, preferably, to the rotatable bonds and evaluates the energy following the change. Metropolis Monte Carlo, and Genetic algorithm are popular stochastic search methods that are used as conformational search tools in docking programs.

- *Monte Carlo*: It works on the principle of random sampling for a predetermined number of iterations to locate optimum conformation. The search algorithm starts by making a random changes in any of randomly selected rotatable bond. Following which, it evaluates the energy and compares it with the starting conformation. If the energy of the new state is lower than the previous state, then the conformation is accepted. If the energy of the new state is higher than the previous state, then it assigns a random number between 0 and 1 and evaluates its probability against the Boltzmann distribution; if the random number assigned is greater than the probability, then the new state is accepted otherwise rejected. In case of the rejection, the algorithm restarts from the previous conformations. Glide[**?**] docking program is also known to employ Monte Carlo simulation in its docking algorithm to improve its pose prediction accuracy.
- *Genetic Algorithm (GA)*: This method[**?**, **?**] is motivated from the process of natural selection, that evaluates energy or docking score as the fitness criteria for selection and breeding. Akin to natural selection, the conformational degrees of freedom are encoded as binary numbers which represents specify values of the torsion angles. At the beginning, several random mutations are made in the binary numbers (interchanging “0” to “1”) and generating an initial population. The fitness score is evaluated for each member. Following which the top ranked (fittest members) are retained for breeding. To generate future generations, either cross-over is performed or mutations. This process is repeated for a pre-determined number of iterations. Autodock[**?**] and GOLD[**?**] are examples of docking programs that use genetic algorithm as their conformational search tool.

Molecular Dynamics Simulations

At this point, it is appropriate to gently introduce another powerful conformational search method, namely *molecular dynamics (MD) simulations*. It simulates the atomic motions as a function of time by solving the Newton's equation of motion[**?**]. MD simulations and molecular docking are complementary[**?**, **?**] that are commonly employed in structure-based drug discovery to understand the protein-ligand interactions and their dynamics aspects. For computational efficiency, molecular docking algorithms treat receptors atoms rigid and the ligand as flexible. This leads to incorrect pose prediction when induced fit binding is observed. Therefore, MD simulation is an elegant method to incorporate such induced fit effects. This can be achieved in two ways, as shown below:

- Pre-docking step to sample various conformations of the receptor without the influence of the ligand. This can serve as multiple receptor conformations for docking.
- Post-docking step to refine the docked receptor-ligand conformation by allowing the complex to evolve to their realistic conformation, which would be more probable physiologically.

Scoring Functions in Molecular Docking

Scoring functions are designed to reproduce the binding thermodynamics [**?**, **?**], *i.e.*, binding enthalpy and entropy (see Eq **??**).

$$\Delta G_{binding} = \Delta H - T\Delta S \tag{0.1}$$

Where ΔH is the enthalpy component of binding and $T\Delta S$ the entropy component of binding at temperature "T". The enthalpy of binding can be estimated by the number and type of interactions between the receptor and ligand at the atomistic level. Hence, scoring functions often estimate the enthalpy component by summing all interactions of different types. However, this is a non-trivial task, which has also been criticised for being treated as purely additive[**?**, **?**]. Nevertheless, different types of scoring functions employ different methods to fit experimental binding data to develop scoring functions. This leads to different types of scoring functions that can be classified into four types. We briefly discuss the scoring functions and their types, for more detailed discussions the readers are directed to consults these papers[**?**, **?**].

For a docking tool to be effective and yield reliable result, the search algorithm and the scoring function are crucial and carefully developed. The single most important question that is often asked about a docking algorithm is "How efficiently can the algorithm differentiate between binders and non-binders?" For the docking algorithm to efficiently differentiate between them, both the conformation search algorithm and the scoring functions must work in tandem at their best capacity. A huge data set of chemical decoys is employed to validate the accuracy, which is spiked with known binders. Chemical decoys are chemical structures that physically resemble known binders but have a different topological arrangements of atoms. The accuracy of the docking algorithm is assessed by its ability to recover known binders from the pool of decoys. However, it is overambitious to expect any docking algorithm to recover all the known binders and none of the decoys. To quantify the accuracy of the docking algorithms and the parameters used, an enrichment factor (EF)[**?**] is used, which is computed using Eq **??**.

$$EF = \frac{\frac{ligands_{(recovered)}}{N_{(recovered)}}}{\frac{ligands_{(total)}}{N_{(total)}}} \tag{0.2}$$

Here, $ligands_{(total)}$ and $ligands_{(recovered)}$ denote the total number of known binders and binders recovered, respectively, and $N_{(total)}$ and $N_{(recovered)}$ denote the total number of decoys and recovered decoys, respectively. From the formula, we can infer that a higher value suggests an effective docking algorithm and parameters thereof. A comprehensive discussion on search algorithms and scoring functions is beyond the scope of this article, and the readers are advised to read the listed papers in the literature [**?**, **?**]. However, we enlist the scoring functions with their succinct descriptions.

Force field-based scoring function:

Force fields[**?**] are mathematical frameworks that describe the potential energy of a molecular system as a function of the positions of its atoms. In other words, they describe chemical information in the numerical information that is more suitable for computations. To do so, various components, each representing a specific type of interaction between atoms. Each component is characterized by its own parameters, such as the strength of the interaction and the range of distances (r) over which it is effective. A force field typically consists of:

1. Bonded Interactions
    a. Bond stretching
    b. Angle bending
    c. Torsional interactions
    d. Improper torsions
2. Non-Bonded Interactions
    a. Van der Waals forces
    b. Electrostatics

The bond stretching and angle bending are generally considered hard terms because the energy required to bring about a small change is generally very large. Moreover, to improve the computational speed, these are treated as "fixed" or "constant". The torsional and improper terms are changed to bring about a conformational change by the conformational search algorithm. Owing to such changes, the non-bonded interactions vary as different conformations lead to different atomic interactions, which is show in Eq **??** as a docking score.

$$E = \sum_i \sum_j \left(\frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^{6}} + \frac{q_i q_j}{\epsilon(r_{ij}) r_{ij}} \right) \tag{0.3}$$

Where r_{ij} is the atomic distance between protein atom i and ligand atom j, A_{ij} and B_{ij} are vdW repulsive and attractive parameters, respectively. q_i and q_j are atomic partial charges on atoms i and j. The first two terms estimate the van der Waals interactions using Lennard-Jones 12–6 potentials, which are commonly used in most force fields. The last term estimates the electrostatic component of the interaction, with solvent effects implicitly defined using the term $\epsilon(r_{ij})$, which is the distance-dependent dielectric constant. It is known that incorporating solvation effects is far from perfect, and without it, the electrostatics will be overestimated for the charged ligands owing to the absence or incorrect dielectric effect seen in the solvated systems. There are several approximations introduced such as Generalised-Born (GB)[**?**] and Poisson-Boltzmann (PB) models [**?**, **?**], which are popular and computationally efficient methods that estimate solvent effects.

Empirical scoring function:

$$\Delta G = \sum_i W_i \cdot \Delta G_{vdW} + \sum_j W_j \cdot \Delta G_{ele} + \sum_k W_{solv} \cdot \Delta G_{solv} + ... \qquad (0.4)$$

Where ΔG_{vdW}, ΔG_{ele}, and ΔG_{solv}, depict the free energy contribution of van der Waals', electrostatic, solvation/desolvation, etc. $\sum_i W_i$, $\sum_j W_j$, and $\sum_k W_{solv}$ are the corresponding coefficients that are adjusted to fit the experimental binding affinity data. The final score is obtained as the sum of all the energy components. Different scoring functions in this class include or exclude certain components, while others use a different fitting method that best fits the binding affinity data.

Knowledge-based scoring function:

$$w(r) = -k_B T [\frac{\rho(r)}{\rho^*(r)}] \qquad (0.5)$$

Where $w(r)$ is the final score which is dependent on the atomic distances r. k_B is the Boltzmann constant, T is the absolute temperature in K. $\rho(r)$ is the frequency of protein-ligand atom pairs at distance r, and $\rho^*(r)$ is the frequency of protein-ligand atom pairs in the reference state at distance r. The reference state is generally the frequency that a particular atom pair is observed in the experimental structures. The type of interaction (intermolecular atom pairs) that appears more frequently in the reference state will be scored higher. Therefore, the ratio $\frac{\rho(r)}{\rho^*(r)}$ can be considered as a pair distribution function.

Consensus scoring function:

$$\Delta G = W_1 \cdot scoringfunction_1 + W_2 \cdot scoringfunction_2 + W_3 \cdot scoringfunction_3 + ... \qquad (0.6)$$

Where $W_{1,2,3,..}$ are weights assigned to each of the $scoringfunction$ 1, 2, 3, etc. These weights may not be directly linked to the atomic physicochemical properties. As we have seen so far, each type of scoring function comes with its own set of advantages and disadvantages. There have been many attempts to use more than one type of scoring function simultaneously to improve their overall scoring accuracy. An ideal consensus scoring function will use one or more scoring functions that complement the pros and cons. This is expected to minimise the cons and amplify the pros of each scoring function. However, this is far from reality, and often times, new limitations are observed.

Few questions that frequently arise, especially among the beginners, is *what are generally the units of the docking score?* To address this question pertaining to the units, the units are generally units of energy, expresses as ”kcal/mol” or ”kJ/mol”. However, it is not unusual to observe scoring functions that yield unitless docking scores. Irrespective, of the units of the docking score, a generally accepted convention is such that a more negative score indicates a better binding score. We would also point out that not always a more negative score indicates “correct” binding conformation (also referred to as “pose”). *Are docking scores obtained from different scoring functions directly comparable?* This is rather difficult to answer holistically because scoring functions could be comparable as they share common philosophy of development or datasets. In our experience, we have always avoided any head-to-head comparison of scores, nevertheless, we compare the rank-ordering given by different scoring functions[**?**].